

Convolution and Correlation

Convolution Sum

- **The Impulse Response (or Unit Sample Response):** The impulse response is defined as the output of an LTI system due to a unit impulse signal input applied at time $n = 0$.

$$x(n) \rightarrow y(n)$$

$$\delta(n) \rightarrow h(n)$$

where $\delta(n)$ is the unit impulse or unit sample signal and $h(n)$ is the unit impulse response of the discrete-time LTI system (DTS).



Convolution Sum

- **Convolution Sum:** Given the impulse response, we determine the output due to an arbitrary input signal by expressing the input as a weighted superposition of time-shifted impulses.
- By linearity and time-invariance, the output signal must be a weighted superposition of time-shifted impulse responses.
- This weighted superposition is termed the *convolution sum* for discrete-time systems.
- Let $y(n)$ be the system response to impulse input $x(n)$.

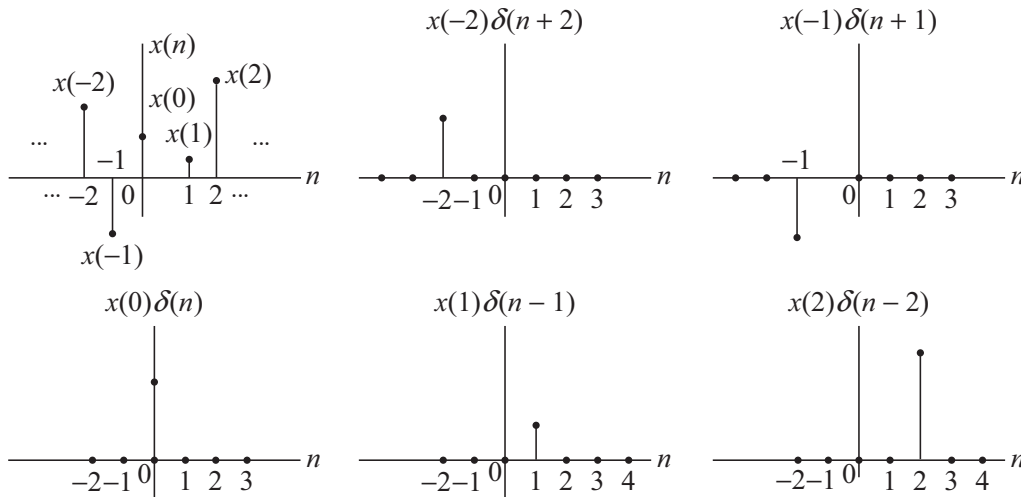
$$x(n) \rightarrow y(n)$$

Convolution Sum contd...

- An arbitrary signal $x(n]$ can be expressed as the weighted sum of time-shifted impulses.

$$x(n) = \cdots + x(-2)\delta(n+2) + x(-1)\delta(n+1) + x(0)\delta(n) + x(1)\delta(n-1) + x(2)\delta(n-2) + \cdots$$

$$x(n) = \sum_{k=-\infty}^{\infty} x(k)\delta(n-k)$$



Convolution Sum contd...

- Using the superposition and time-invariant property of LTI systems, we can express output $y(n)$ as a linear combination of the responses of the system to shifted impulse signals.

$$\delta(n) \longrightarrow h(n)$$

$$\delta(n - k) \longrightarrow h(n - k)$$

$$x(k)\delta(n - k) \longrightarrow x(k)h(n - k)$$

$$x(n) = \sum_{k=-\infty}^{\infty} x(k)\delta(n - k) \longrightarrow y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n - k)$$

- The following relationship is called the *convolution sum* of sequences $x(n)$ and $h(n)$.

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n - k) = x(n) * h(n)$$

Example

Example: Compute the convolution $y(n) = x(n) * h(n)$ of the following pairs of signals:

(a) $x(n) = h(n) = u(n)$

(b) $x(n) = (0.8)^n u(n)$ and $h(n) = (0.4)^n u(n)$

(c) $x(n) = h(n) = a^n u(n)$

Solution: (a) Consider the given signals $x(n) = h(n) = u(n)$. By definition

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k) = \sum_{k=-\infty}^{\infty} u(k)u(n-k)$$

The lower limit on the convolution sum simplifies to $k = 0$ [because $u(k) = 0, k < 0$], the upper limit to $k = n$ [because $u(n-k) = 0, k > n$], and we get

$$y(n) = \sum_{k=0}^n 1 = (n+1)u(n) = r(n+1)$$

Example contd...

(b) Consider the given signals $x(n) = (0.8)^n u(n)$ and $h(n) = (0.4)^n u(n)$. By definition

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k) = \sum_{k=-\infty}^{\infty} (0.8)^k u(k)(0.4)^{n-k} u(n-k)$$

The lower limit on the convolution sum simplifies to $k = 0$ [because $u(k) = 0, k < 0$], the upper limit to $k = n$ [because $u(n-k) = 0, k > n$], and we get

$$\begin{aligned} y(n) &= \sum_{k=0}^n (0.8)^k (0.4)^{n-k} = (0.4)^n \sum_{k=0}^n 2^k \\ &= (0.4)^n \frac{1 - 2^{n+1}}{1 - 2} \\ &= (0.4)^n (2^{n+1} - 1) \\ &= [2(0.8)^n - (0.4)^n] u(n) \end{aligned}$$

Example contd...

(c) Consider the given signals $x(n) = h(n) = a^n u(n)$. By definition

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k) = \sum_{k=-\infty}^{\infty} a^k u(k) a^{n-k} u(n-k)$$

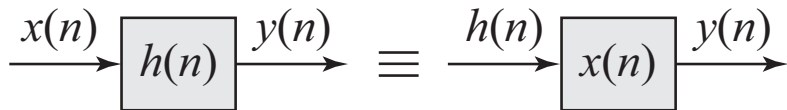
The lower limit on the convolution sum simplifies to $k = 0$ [because $u(k) = 0, k < 0$], the upper limit to $k = n$ [because $u(n-k) = 0, k > n$], and we get

$$\begin{aligned} y(n) &= \sum_{k=0}^n a^k a^{n-k} \\ &= a^n \sum_{k=0}^n 1 \\ &= (n+1)a^n u(n) \end{aligned}$$

Properties of Convolution Sum

- **Commutative Property:** The convolution operation possesses the commutative property, that is,

$$x(n) * h(n) = h(n) * x(n)$$



This property, as shown in Figure, implies that the role of the input signal and impulse response are interchangeable.

- **Proof:** By definition

$$x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

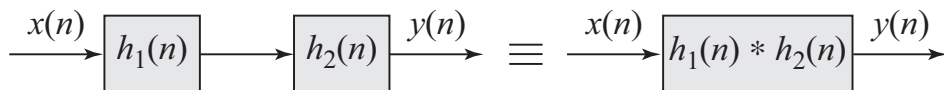
Properties of Convolution Sum

A change of variables is performed by letting $n - k = m$, which also yields $k = n - m$, $m = -\infty$ as $n = -\infty$, and $m = \infty$ as $n = \infty$. Therefore,

$$\begin{aligned}x(n) * h(n) &= \sum_{m=-\infty}^{\infty} x(n - m)h(m) \\&= \sum_{m=-\infty}^{\infty} h(m)x(n - m) \\&= h(n) * x(n)\end{aligned}$$

🔴 **Associative Property:** The convolution sum possesses the associativity, that is,

$$x(n) * [h_1(n) * h_2(n)] = [x(n) * h_1(n)] * h_2(n)$$



Properties contd...

Associativity, as shown in Figure, implies that a cascade combination of LTI systems can be replaced by a single system whose impulse response is the convolution of the individual impulse responses.

Proof: By definition

$$\begin{aligned}x(n) * [h_1(n) * h_2(n)] &= x(n) * \underbrace{\left[\sum_{k=-\infty}^{\infty} h_1(k)h_2(n-k) \right]}_{=h(n)} \\&= x(n) * h(n) = \sum_{m=-\infty}^{\infty} x(m)h(n-m) \\&= \sum_{m=-\infty}^{\infty} x(m) \left[\sum_{k=-\infty}^{\infty} h_1(k)h_2(n-m-k) \right] \\&= \sum_{m=-\infty}^{\infty} x(m) \left[\sum_{k=-\infty}^{\infty} h_1(k)h_2[n-(m+k)] \right]\end{aligned}$$

Properties contd...

A change of variables is performed by letting $r = m + k$, which also yields $k = r - m$, $r = -\infty$ as $k = -\infty$, and $r = \infty$ as $k = \infty$. Therefore

$$x(n) * [h_1(n) * h_2(n)] = \sum_{m=-\infty}^{\infty} x(m) \left[\sum_{r=-\infty}^{\infty} h_1(r - m) h_2(n - r) \right]$$

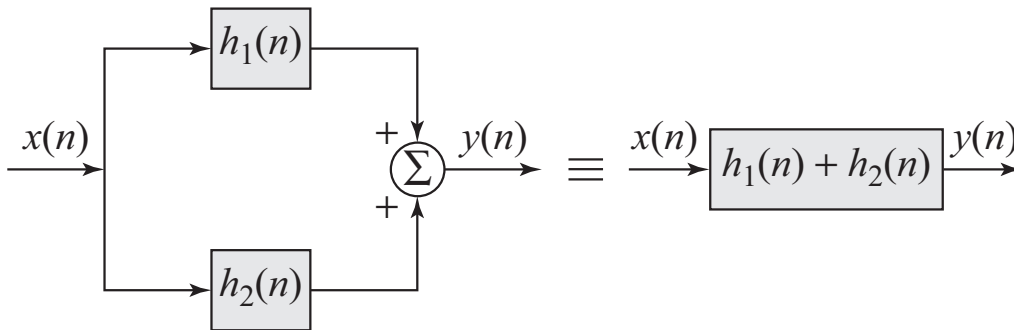
Interchanging the order of summation, gives the desired result

$$\begin{aligned} x(n) * [h_1(n) * h_2(n)] &= \sum_{r=-\infty}^{\infty} \left[\sum_{m=-\infty}^{\infty} x(m) h_1(r - m) \right] h_2(n - r) \\ &= \sum_{r=-\infty}^{\infty} [x(r) * h_1(r)] h_2(n - r) \\ &= [x(n) * h_1(n)] * h_2(n) \end{aligned}$$

Properties contd...

- **Distributive Property:** The convolution operation possesses the distributivity; that is

$$x(n) * [h_1(n) + h_2(n)] = [x(n) * h_1(n)] + [x(n) * h_2(n)]$$



Properties contd...

Distributivity, as shown in Figure, states that a parallel combination of LTI systems is equivalent to a single system whose impulse response is the sum of the individual impulse responses in the parallel configuration.

Proof: By definition

$$\begin{aligned}x(n) * [h_1(n) + h_2(n)] &= \sum_{k=-\infty}^{\infty} x(k)[h_1(n-k) + h_2(n-k)] \\&= \sum_{k=-\infty}^{\infty} x(k)h_1(n-k) + \sum_{k=-\infty}^{\infty} x(k)h_2(n-k) \\&= [x(n) * h_1(n)] + [x(n) * h_2(n)]\end{aligned}$$

Properties contd...

● **Shift Property:** According to the shift property,

if $x(n) * h(n) = y(n)$ then $x(n) * h(n - n_o) = x(n - n_o) * h(n) = y(n - n_o)$

and

$$x(n - n_1) * h(n - n_2) = y(n - n_1 - n_2)$$

● **Proof:** By definition

$$x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k)h(n - k) = y(n)$$

$$x(n) * h(n - n_o) = \sum_{k=-\infty}^{\infty} x(k)h(n - n_o - k) = y(n - n_o)$$

Properties contd...

- **Convolution with an Impulse:** Convolution of a signal $x(n)$ with a unit impulse signal results in the signal $x(n)$ itself.

$$x(n) * \delta(n) = x(n)$$

- **Proof:** By definition

$$x(n) * \delta(n) = \sum_{k=-\infty}^{\infty} x(k)\delta(n-k)$$

From the sampling property of the impulse signal, we have $x(k)\delta(n-k) = x(n)\delta(n-k)$, therefore

$$x(n) * \delta(n) = \sum_{k=-\infty}^{\infty} x(n)\delta(n-k) = x(n) \underbrace{\sum_{k=-\infty}^{\infty} \delta(n-k)}_{=1 \text{ for all } n} = x(n)$$

Properties contd...

- **Width Property:** Let N_x and N_h be the length of the sequences $x(n)$ and $h(n)$ respectively. The length of the sequence $y(n) = x(n) * h(n)$ is

$$N_y = N_x + N_h - 1$$

- **Proof:** Let LE_y , LE_x , and LE_h be the left edge of $y(n)$, $x(n)$ and $h(n)$ respectively and $LE_y = LE_x + LE_h$. Let RE_y , RE_x , and RE_h be the right edge of $y(n)$, $x(n)$ and $h(n)$ respectively and $RE_y = RE_x + RE_h$. The length of the sequences $x(n)$ and $h(n)$ are

$$N_x = RE_x - LE_x + 1 \qquad N_h = RE_h - LE_h + 1$$

The length of the sequence $y(n) = x(n) * h(n)$ is

$$\begin{aligned} N_y &= RE_y - LE_y + 1 \\ &= RE_x + RE_h - LE_x - LE_h + 1 \\ &= (RE_x - LE_x + 1) + (RE_h - LE_h + 1) - 1 \\ N_y &= N_x + N_h - 1 \end{aligned}$$

Properties contd...

● **Sum Property:** Let the sum of all the impulses in the signals $y(n)$, $x(n)$, and $h(n)$ be S_y , S_x , and S_h , respectively. Then

$$S_y = \sum_{n=-\infty}^{\infty} y(n) = \sum_{n=-\infty}^{\infty} x(n) \sum_{n=-\infty}^{\infty} h(n) = S_x S_h$$

● **Proof:** Consider the sum

$$S_y = \sum_{n=-\infty}^{\infty} y(n) = \sum_{n=-\infty}^{\infty} [x(n) * h(n)] = \sum_{n=-\infty}^{\infty} \left[\sum_{k=-\infty}^{\infty} x(k)h(n-k) \right]$$

interchanging the order of the summation,

$$S_y = \sum_{k=-\infty}^{\infty} x(k) \sum_{n=-\infty}^{\infty} h(n-k) = \underbrace{\sum_{k=-\infty}^{\infty} x(k)}_{S_x} \underbrace{\sum_{m=-\infty}^{\infty} h(m)}_{S_h} = S_x S_h$$

Relations between LTI system Properties and the Impulse Response

LTI Systems with and without Memory: A discrete-time LTI system is said to be memoryless if and only if the impulse response

$$h(n) = K\delta(n)$$

where $K = h(0)$ is a constant.

Proof: The output of a discrete-time LTI system with impulse response $h(n)$ is given by

$$\begin{aligned} y(n) &= h(n) * x(n) = \sum_{k=-\infty}^{\infty} h(k)x(n-k) \\ &= \underbrace{\cdots + h(-2)x(n+2) + h(-1)x(n+1)}_{=0} + h(0)x(n) + \underbrace{h(1)x(n-1) + h(2)x(n-2) + \cdots}_{=0} \end{aligned}$$

For this system to be memoryless, $y(n)$ must depend only on $x(n)$. This implies that $h(k) = 0$ for $k \neq 0$; thus an LTI system is memoryless if and only if

$$h(n) = K\delta(n)$$

LTI system Properties contd...

Causality for LTI Systems: A discrete-time LTI system is said to be causal if and only if

$$h(n) = 0 \quad n < 0$$

● **Proof:** The output of a discrete-time LTI system with impulse response $h(n)$ is given by

$$y(n) = h(n) * x(n) = \sum_{k=-\infty}^{\infty} h(k)x(n-k) = \left[\sum_{k=-\infty}^{-1} h(k)x(n-k) \right] + \left[\sum_{k=0}^{\infty} h(k)x(n-k) \right]$$
$$y(n) = \underbrace{[\cdots + h(-2)x(n+2) + h(-1)x(n+1)]}_{=0} + [h(0)x(n) + h(1)x(n-1) + h(2)x(n-2) + \cdots]$$

For a causal system $y(n)$ depends only on past and present values of the input. Hence,

$$h(n) = 0 \quad \text{for } n < 0$$

LTI system Properties contd...

- **Stability for LTI Systems:** A discrete-time LTI system is said to be stable if and only if its impulse response is absolutely summable, i.e.,

$$\sum_{n=-\infty}^{\infty} |h(n)| < \infty$$

- **Proof:** The output of a discrete-time LTI system is given by

$$y(n) = h(n) * x(n)$$

$$y(n) = \sum_{k=-\infty}^{\infty} h(k)x(n-k)$$

$$|y(n)| = \left| \sum_{k=-\infty}^{\infty} h(k)x(n-k) \right|$$

$$|y(n)| \leq \sum_{k=-\infty}^{\infty} |h(k)||x(n-k)|$$

LTI system Properties contd...

If we assume that the input is bounded, or $|x(n)| \leq B_x < \infty$, then $|x(n - k)| \leq B_x$, and it follows that

$$|y(n)| \leq B_x \underbrace{\sum_{k=-\infty}^{\infty} |h(k)|}_{=B_y}$$
$$|y(n)| \leq B_x B_y$$

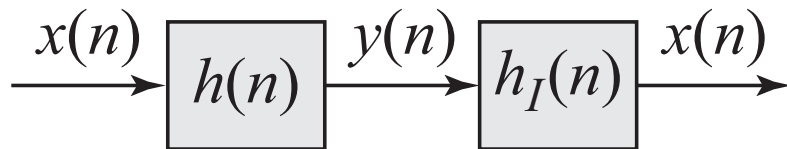
From above equation, we can conclude that if the impulse response is *absolutely summable*, that is, if

$$\sum_{k=-\infty}^{\infty} |h(k)| = B_y < \infty$$

then $y(n)$ is bounded in magnitude, and hence, the system is stable.

LTI system Properties contd...

- **Invertibility for LTI Systems:** Consider a discrete-time LTI system with impulse response $h(n)$. This system is invertible only if an inverse system exists that, when connected in series with the original system, produces an output equal to the input to the first system. Furthermore, if an LTI system is invertible, then it has an LTI inverse.



- The impulse response of the cascade connection shown in Figure, is the convolution of $h(n)$ and $h_I(n)$. We require the output of the cascade to equal the input, or

$$x(n) * [h(n) * h_I(n)] = x(n)$$

This requirement implies that

$$h(n) * h_I(n) = \delta(n)$$

LTI system Properties contd...

- **The Unit Step Response of an LTI Systems:** The step response $s(n)$ is defined as the output due to a unit step input signal $x(n) = u(n)$. The step response of a discrete-time LTI system with impulse response $h(n)$ is given by

$$s(n) = \sum_{k=-\infty}^n h(k)$$

- **Proof:** The step response of a discrete-time LTI system is the convolution of the unit step with the impulse response; that is,

$$s(n) = h(n) * u(n) = \sum_{k=-\infty}^{\infty} h(k)u(n-k)$$

Now, since

$$u(n-k) = \begin{cases} 1 & k \leq n \\ 0 & k > n \end{cases}$$

LTI system Properties contd...

we have

$$s(n) = \sum_{k=-\infty}^n h(k)$$

From this equation, it is clear that $h(n)$ can be recovered from $s(n)$.

$$s(n) = h(n) + \sum_{k=-\infty}^{n-1} h(k)$$

$$s(n) = h(n) + s(n-1)$$

$$h(n) = s(n) - s(n-1)$$

The step response of a discrete-time LTI system is the running sum of its impulse response. Conversely, the impulse response of a discrete-time LTI system is the first difference of its step response.

Correlation of Signals

- **Correlation of Signals:** Correlation between the two signals is to measure the degree to which the two signals are similar. It closely resembles convolution.

Correlation of Signals

- **Correlation of Signals:** Correlation between the two signals is to measure the degree to which the two signals are similar. It closely resembles convolution.
- Correlation finds widespread use in applications such as target ranging (radar, sonar), digital communication (correlation receiver) etc.

Correlation of Signals

- **Correlation of Signals:** Correlation between the two signals is to measure the degree to which the two signals are similar. It closely resembles convolution.
- Correlation finds widespread use in applications such as target ranging (radar, sonar), digital communication (correlation receiver) etc.
- **Crosscorrelation Sequence of Discrete-Time Energy Signals:** Let $x(n)$ and $y(n)$ be two real finite energy signals. The crosscorrelation of $x(n)$ and $y(n)$ is a signal $R_{xy}(m)$, which is defined as

$$R_{xy}(m) = \sum_{n=-\infty}^{\infty} x(n)y(n-m) = \sum_{n=-\infty}^{\infty} x(n+m)y(n)$$

or,

$$R_{yx}(m) = \sum_{n=-\infty}^{\infty} y(n)x(n-m) = \sum_{n=-\infty}^{\infty} y(n+m)x(n)$$

By comparing two equations, we conclude that

$$R_{xy}(m) = R_{yx}(-m)$$

Correlation contd...

- For energy signals, there is a simple mathematical relationship between correlation and convolution sum,

$$R_{xy}(m) = x(m) * y(-m) = \sum_{n=-\infty}^{\infty} x(n)y(-(m-n)) = \sum_{n=-\infty}^{\infty} x(n)y(n-m)$$

- Crosscorrelation Sequence of Power Signals:** The crosscorrelation function between two real power signals $x(n)$ and $y(n)$ is defined by

$$R_{xy}(m) = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x(n)y(n-m) = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x(n+m)y(n)$$

or

$$R_{xy}(m) = \frac{1}{2N+1} \sum_{n=-N}^N x(n)y(n-m)$$

Correlation contd...

- **Autocorrelation Sequence of Discrete-time Signals:** In the special case where $y(n) = x(n)$, we have the autocorrelation sequence of $x(n)$. If $x(n)$ is a real energy signal, its autocorrelation is

$$R_{xx}(m) = \sum_{n=-\infty}^{\infty} x(n)x(n-m) = \sum_{n=-\infty}^{\infty} x(n+m)x(n)$$

There is a simple mathematical relationship between correlation and convolution sum,

$$R_{xx}(m) = x(m) * x(-m) = \sum_{n=-\infty}^{\infty} x(n)x(-(m-n)) = \sum_{n=-\infty}^{\infty} x(n)x(n-m)$$

Correlation contd...

- **Autocorrelation Sequence of Discrete-time Signals:** In the special case where $y(n) = x(n)$, we have the autocorrelation sequence of $x(n)$. If $x(n)$ is a real energy signal, its autocorrelation is

$$R_{xx}(m) = \sum_{n=-\infty}^{\infty} x(n)x(n-m) = \sum_{n=-\infty}^{\infty} x(n+m)x(n)$$

There is a simple mathematical relationship between correlation and convolution sum,

$$R_{xx}(m) = x(m) * x(-m) = \sum_{n=-\infty}^{\infty} x(n)x(-(m-n)) = \sum_{n=-\infty}^{\infty} x(n)x(n-m)$$

- If $x(n)$ is a power signal, its autocorrelation is given by

$$R_{xx}(m) = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x(n)x(n-m) = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x(n+m)x(n)$$

Correlation contd...

- The autocorrelation of a periodic signal $x(n)$ (all periodic signals are power signals) with period N is given by

$$R_{xx}(m) = \frac{1}{2N+1} \sum_{n=-N}^N x(n)x(n-m)$$

Correlation contd...

- The autocorrelation of a periodic signal $x(n)$ (all periodic signals are power signals) with period N is given by

$$R_{xx}(m) = \frac{1}{2N+1} \sum_{n=-N}^N x(n)x(n-m)$$

- **Properties of Crosscorrelation and Autocorrelation Sequences:**

- The crosscorrelation sequence satisfies the property

$$R_{xy}(m) = R_{yx}(-m)$$

The autocorrelation sequence is an even function.

$$R_{xx}(m) = R_{xx}(-m)$$

Correlation contd...

- Relation to signal energy and signal power:

Case I: If $x(n)$ is an energy signal

$$R_{xx}(0) = E_x = \sum_{n=-\infty}^{\infty} x^2(n)$$

Case II: If $x(n)$ is a power signal

$$R_{xx}(0) = P_x = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x^2(n)$$

- The crosscorrelation sequence satisfies the condition

$$|R_{xy}(m)| \leq \sqrt{R_{xx}(0)R_{yy}(0)} = \sqrt{E_x E_y}$$

and the autocorrelation sequence satisfies this condition

$$|R_{xx}(m)| \leq R_{xx}(0) = E_x$$

Convolution and Correlation

Thank You!